



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 8 Verification of Thevenin's Theorem

SUBMITTED TO:

Mr. Mehmood-ul-Haq

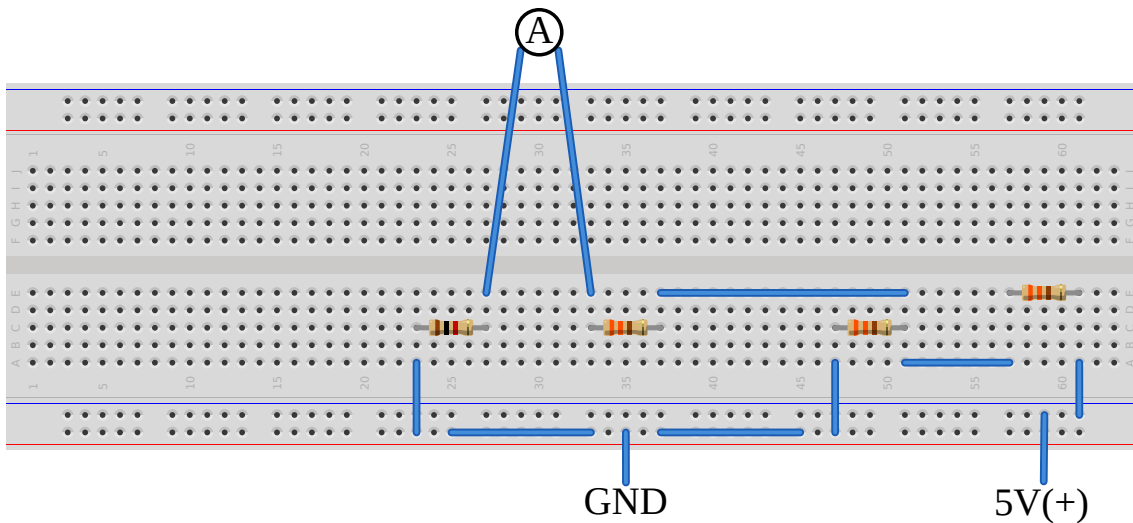
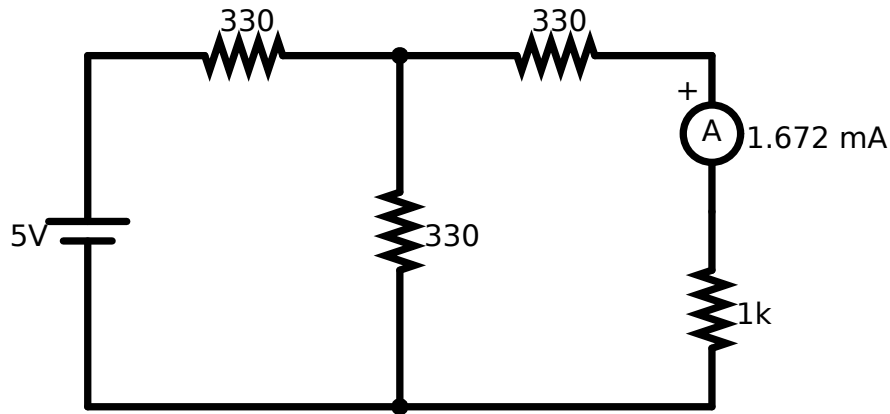
SUBMITTED BY:

Muhammad Saad (2025-CE-113)

Submission Date: 12 May, 2026

Department of Computer Engineering
University of Engineering and Technology Lahore
Main Campus

Experiment 1



Breadboard circuit layout: Initial assembly with load resistance $R_L = 1\text{ k}\Omega$

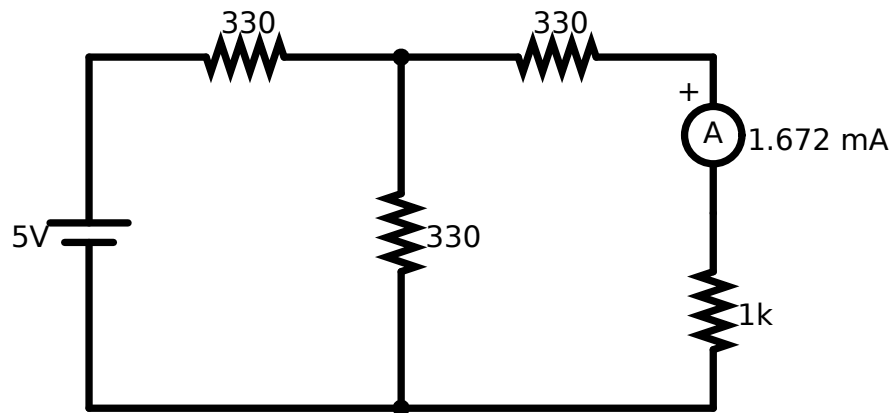
Practical Procedure

The objective of this experiment is to verify Thevenin's Theorem by simplifying a complex resistive network into an equivalent circuit featuring a single voltage source (V_{TH}) in series with an equivalent resistance (R_{TH}). The primary circuit is constructed using three $330\text{ }\Omega$ resistors and a $1\text{ k}\Omega$ load resistor (R_L), driven by a Regulated Power Supply. For all measurements, the **UNI-T UT89XE** digital multimeter is utilized to capture the load current, open-circuit voltage, and

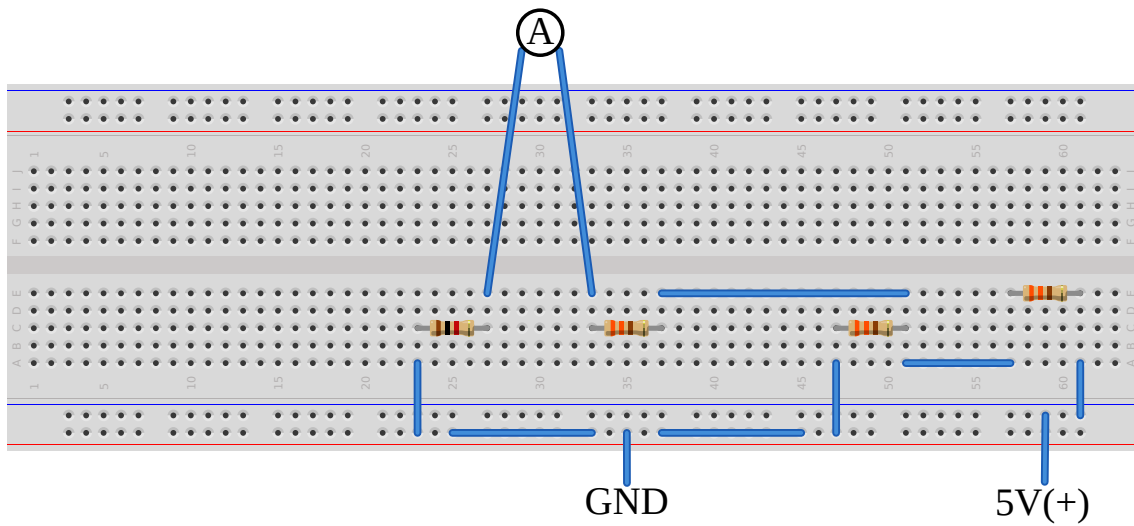
looking-back resistance. We provide supply voltage of 5V from the RPS to the initial circuit to observe current while load is connected.

To measure the load current (I_L), the multimeter dial is set to the **20mA DC** range (indicated by the **A—** symbol). The black probe is connected to the **COM** port while the red probe is inserted into the **mA μ A** terminal. This measurement requires the DMM to be connected in **series** with the load. To find the Thevenin voltage (V_{TH}), the load resistor is removed, the multimeter is switched to the **20V** or **200V DC** range (**V—** symbol), and probes are placed in **parallel** across the open terminals. To determine the Thevenin resistance (R_{TH}), the RPS is removed and short-circuited; the DMM is then set to the **2000 Ω** range and probes are connected across the load terminals to measure the "looking-back" resistance.

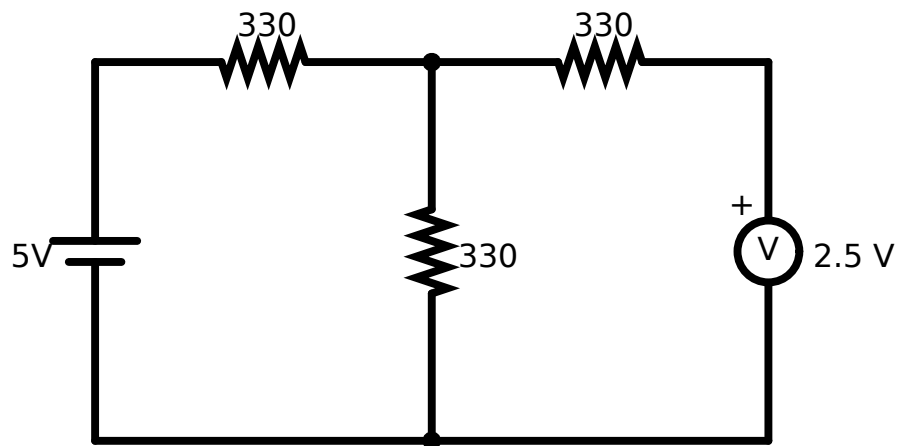
After obtaining the values for V_{TH} and R_{TH} , the Thevenin equivalent circuit is constructed by setting the RPS to the measured V_{TH} and placing a potentiometer adjusted to the R_{TH} value in series with the 1 k Ω load resistor. The final objective is to verify whether the equivalent circuit produces the same load current (I_L) as recorded in the original network. Theoretical accuracy depends on the linearity of the components; however, minor discrepancies may be observed in simulated or physical environments. Factors such as the **internal resistance** of the power supply and the **burden voltage** of the ammeter can introduce slight deviations between the mathematical predictions and the recorded experimental data.



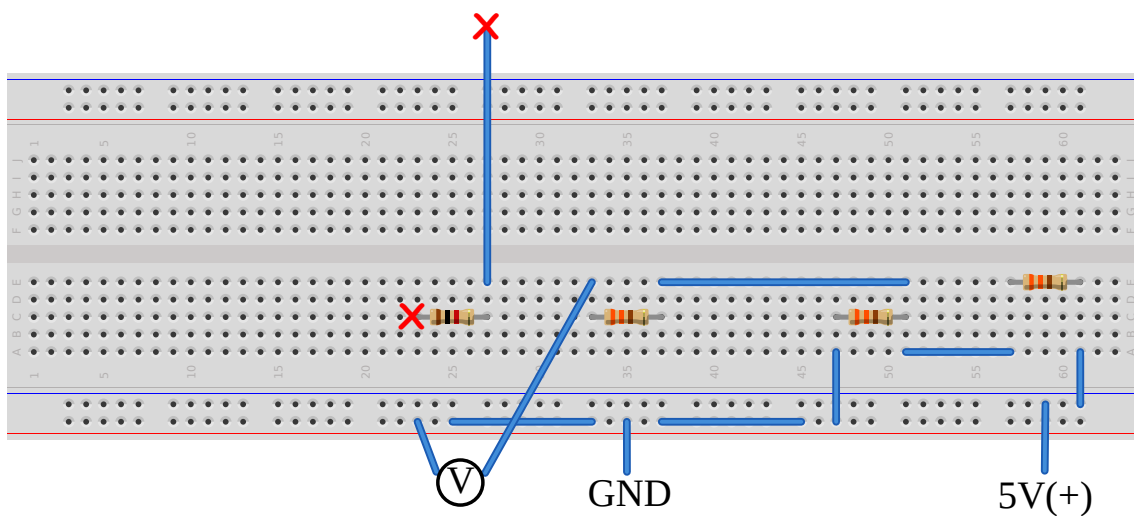
Falstad: Measurement of Load Current I_L in the original network



Breadboard: Series ammeter connection for I_L measurement

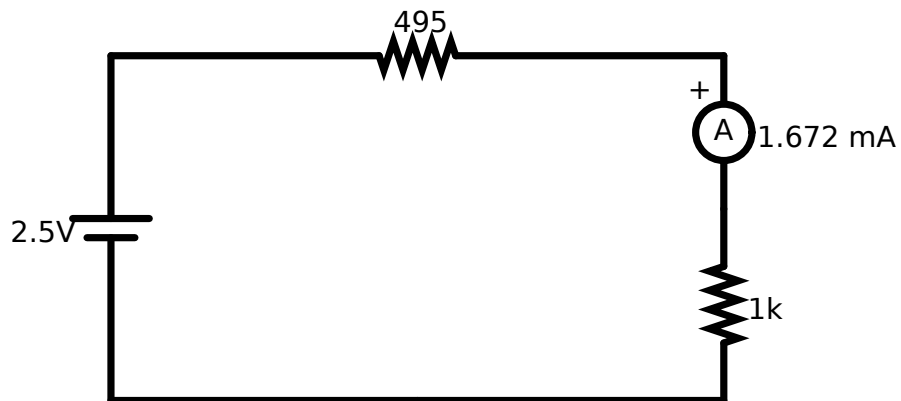


Falstad: Open-circuit voltage measurement for V_{TH}

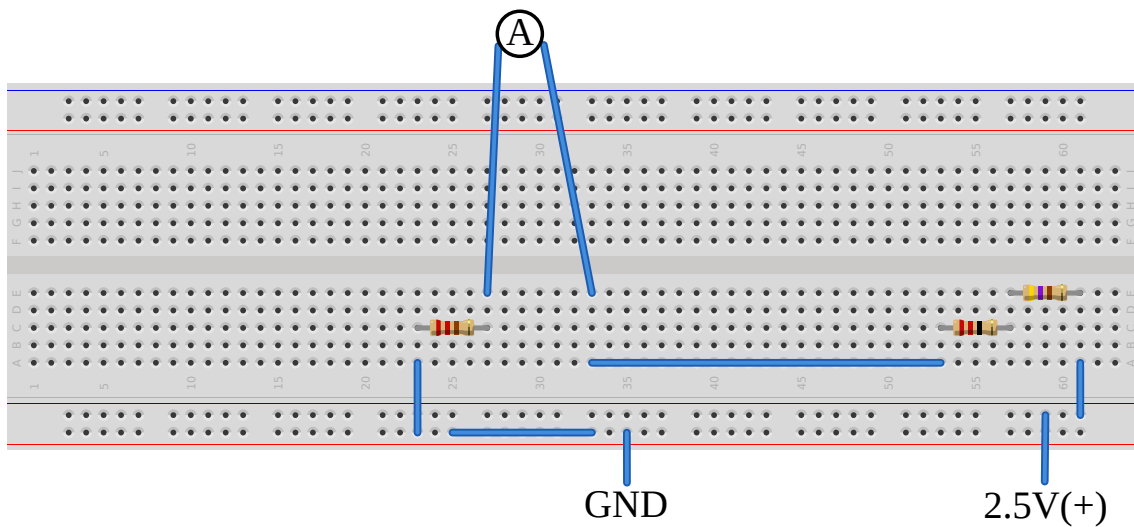


Breadboard: Parallel voltmeter connection for V_{TH} (Load removed)

Breadboard: Ohmmeter connection for R_{TH} calculation



Falstad: Verification of I_L using the Thevenin Equivalent Circuit



Breadboard: Equivalent circuit using RPS (V_{TH}) and a series combination approximating R_{TH} . A dual-resistor network was utilized as a practical substitute for a potentiometer.

Experiment 1 Data

Practical Values

Source E	V_{TH}	R_{TH}	Load Current I_L	
			Original Circuit	Equivalent Circuit
V	V	Ω	mA	mA
5.0	2.5	495	1.672	1.672

Post-Lab Questions

1. What is Thevenin's Theorem?

Thevenin's Theorem states that any linear, bilateral, active two-terminal network can be replaced by an equivalent voltage source (V_{TH}) in series with a looking-back resistance R_{TH} . V_{TH} is defined as the open-circuit voltage (V_{OC}) across the terminals when the load is removed.

2. What are the applications of Thevenin's Theorem?

Thevenin's Theorem is widely used for circuit simplification, particularly in determining the full load current for varying loads without re-analyzing the entire network. It is essential in power system analysis, modeling battery sources with internal resistance, and electronic design to calculate input and output impedances of amplifier stages.

3. What are the theoretical and practical limitations of Thevenin's Theorem?

Theoretically, the theorem is restricted to linear and bilateral networks; it cannot be applied directly to circuits with non-linear elements like diodes or transistors. Practically, limitations include the power dissipation limits of equivalent components and the difficulty in measuring R_{TH} in active circuits containing dependent sources or complex impedances in high-frequency AC applications.